EvalúaYa — Algorithm technical specification

How the app decides Green, Yellow, Orange or Red. Every statement is taken from the working code (the source file is noted in grey).

How the final color is decided

The result is formed in two independent layers and the more severe one is kept. The safety rules (Layer 1) can only raise the level — they never make a result less severe than the AI suggested.

Order: read the earthquake data at the location (ShakeMap) -> photos and answers go to the AI for a Green/Yellow/Orange/Red call -> apply the deterministic rules -> combine by taking the worse of the two.

The four levels: **Green** no significant structural damage (appears safe to occupy); **Yellow** light or cosmetic damage (habitable, keep monitoring); **Orange** moderate-to-serious damage that needs an engineer soon (limit use to short entries); **Red** serious damage or signs of collapse (do not enter, evacuate).

source: assessment.functions.ts — finalRisk = maxRisk(ai, rules)

Layer 1 — Deterministic safety rules

Same inputs, same result, no AI. Each group sets a severity floor; the highest one that fires is kept (and the AI can never lower it).

Condition -> RED	Why
YES to ground-liquefaction signs.	Soil loses bearing capacity; settlement or tilting possible.
YES to pounding against a neighboring building.	Impact can cause severe, hidden structural damage.
YES to severe plumbing / gas damage.	Possible gas leak: an immediate life-safety hazard.
Strong shaking **AND** any reported structural damage.	MMI ≥ 8 or PGA ≥ 0.50g together with visible damage = critical combo.
Unreinforced masonry (URM) **with** structural damage or moderate shaking (MMI ≥ 6 or PGA ≥ 0.25g).	With no steel reinforcement, damage or shaking makes it unsafe to enter.

Condition -> ≥ ORANGE	Why
Unreinforced masonry (URM) alone, no visible damage or strong shaking.	Because it is so brittle it needs a professional look soon.
Severe shaking: MMI ≥ 8 **or** PGA ≥ 0.50g (even with no visible damage).	This height/location was shaken hard; review soon.
Spectral demand ≥ 0.40g at the building's period.	Buildings of that height felt the shaking especially hard.
Very soft soil (very low vs30).	Strongly amplifies shaking and raises liquefaction risk.

Condition -> ≥ YELLOW	Why
Moderate shaking: MMI ≥ 6 **or** PGA ≥ 0.25g.	Appreciable shaking; inspect more carefully even with no visible damage.
Soft soil (low vs30).	Amplifies shaking and raises liquefaction / settlement risk.
More than 7 floors.	Greater consequence; upper floors need professional review.
Structural system CMF, CIW, PCF or RML.	Concrete / reinforced-masonry systems warrant extra caution.

If none apply, Layer 1 contributes Green and the AI's decision stands.

source: safety-rules.ts — evaluateSafetyRules()

USGS ShakeMap data (the data-driven layer)

If the resident shares their location, we read the official USGS ShakeMap grid for the active earthquake. The grid carries several co-registered layers, bilinearly interpolated to the building's exact point:

Layer	What it contributes
MMI (Mercalli intensity)	How strongly shaking was felt (scale I–XII).
PGA — peak acceleration (g)	Max instantaneous shaking; thresholds 0.25g and 0.50g.
PGV — peak velocity (cm/s)	Shaking energy; context for the AI.
SA(0.3 / 0.6 / 1.0 / 3.0 s)	Spectral acceleration by period: demand by height.
vs30 -> soil class	Rock / stiff / soft / very soft (amplification, liquefaction).

Demand by height. We estimate the building's natural period as T ≈ 0.1 × number of floors and pick the nearest spectral band (≤0.45s->SA0.3; ≤0.8s->SA0.6; ≤2.0s->SA1.0; else->SA3.0). That band's value is the shaking a building of that height actually experienced, and it feeds both the spectral-demand rule (≥0.40g) and the AI context.

Soil. vs30 ≥ 760 = rock; ≥ 360 = stiff; ≥ 180 = soft; below = very soft. Soft soil raises the caution level because it amplifies shaking.

source: shakemap.ts — seismicAt(), spectralDemand(), buildingPeriod(), soilClassFromVs30()

Layer 2 — AI photo triage

Photos and answers go to a vision model (google/gemini-2.5-flash) acting as a structural engineer doing rapid post-earthquake triage (ATC-20 style), which picks exactly one level:

Level	Meaning
GREEN	No significant structural damage; appears safe to occupy.

YELLOW	Light or cosmetic damage; habitable, keep monitoring and address the items noted.
ORANGE	Moderate-to-serious damage; needs an engineer soon, limit use to short entries.
RED	Serious damage or signs of collapse; unsafe, evacuate immediately.

The ground-motion context (MMI, PGA, PGV, period-matched spectral demand and soil class) is included in the prompt: strong shaking makes any reported damage weigh more, but shaking alone, with no observed damage, does not by itself force Red. The AI is conservative: when safety is uncertain, it does not choose green. It reserves Yellow for genuinely minor issues; if structural elements are affected but collapse is not imminent, it chooses Orange.

source: assessment.functions.ts — SYSTEM_PROMPT + buildPrompt()

How it combines — examples

AI says	Rule fired?	Final result
Yellow	Liquefaction = YES	**RED**
Green	PGA 0.55g + crack = YES	**RED**
Green	URM + crack = YES	**RED**
Green	URM alone (no damage)	**ORANGE**
Yellow	Spectral demand 0.45g	**ORANGE**
Green	9 floors	**YELLOW**
Green	PGA 0.30g (no damage)	**YELLOW**
Green	No rule fired	**GREEN**

source: safety-rules.ts — maxRisk(a, b) returns the more severe

Sources and limits

- ATC-20 (Applied Technology Council): rapid post-earthquake inspection method.
- USGS ShakeMap: official ground-motion grid for the active earthquake.
- Unreinforced masonry (URM) vulnerability: established seismic literature.
- AI vision model: google/gemini-2.5-flash via Lovable AI Gateway.

Limits: this is preliminary guidance, not a certification. It only evaluates the visible items you report; it does not inspect hidden elements or inside walls. It does not replace a licensed structural engineer or Civil Protection.